

Chapter 10

Measuring Trust in Human Robot Interactions: Development of the “*Trust Perception Scale-HRI*”

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10.1 Introduction

Robotics technology has vastly advanced in recent years leading to improved functional capability, robust intelligence, and autonomy of the system. However, along with the added benefits of these technical advancements also come changes in the way in which humans will use or interact with the system. The most prevalent change can be seen in the vision for robot design and development for future human-robot interaction (HRI). This vision is now directed toward a greater prevalence of robotic technologies in context-driven tasks that require social-based interaction. More specifically, robots are beginning to shed their more passive tool-based roles and move more towards being an active integrated team member (Chen and Terrence 2009). The intricacies of HRI are bound to change in order to accommodate the integration of a robot as it becomes more of a companion, friend, or teammate, rather than strictly a machine. Thus, this change in direction has a direct translation of the human role as less of an operator and more of a team member or even a bystander. Thereby, the individual’s trust in that robot takes a prominent role in the success of any interaction, including the future use of said robot.

This chapter works in conjunction with the rest of the book such that the prevalent focus is on the intersection of robust intelligence (RI) and trust in robotic systems. Therefore, to reduce redundancy, we will limit the background to point to the difficulties relating to trust development and the potential issues that occur when trust is not developed appropriately. First, it is important to note that there are a number of factors that influence trust development. Large scale literature reviews of both HRI and human-automation interaction point to the importance of trust

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antecedents relating to the human, the robot, and the environment (Hancock et al. 2011; Schaefer et al. 2014). The key findings from the associated meta-analyses point to the fact there is still much to learn about how trust develops. However, what is prevalent in the literature is the finding that until trust between a human and a robot is solidly established, robotic partners will continue to be underutilized or unused, therefore providing little to no opportunity for trust to develop in the first place (Lussier et al. 2007). This is in part due to the fact that one of the most significant challenges for successful collaboration between humans and robots is the development of appropriate levels of mutual trust in robots (Desai et al. 2009; Groom and Nass 2007). So, regardless of the domain of application, the environment, or the task, a human's trust in their non-human collaborator is an essential element required to ensure that any functional relationship will ultimately be effective.

Research has continued to address the creation and validation of successful evaluation methods for a wide spectrum of HRI issues, including this issue of human-robot trust (Steinfeld et al. 2006). Yet, a limitation in the field has been related to accurate measurement of trust specific to the unique nature of HRI. Human-robot trust is currently measured through subjective assessment. However, these previous studies have been limited by using measurement tools that are a single self-report item (e.g., How much do you trust this robot?) or are an adapted human-interpersonal or human-automation trust scale. The concern with this methodology is that neither of those options truly assesses the full scope of human-robot trust, and brings to question the accuracy of the trust scores. There has been one notable exception: Yagoda and Gillan (2012) developed a subjective human-robot trust scale that is specific to military application. However, the changing vision of HRI continues to press the inclusion of robotic technologies into multiple contextual domains that incorporate varying levels of autonomy, intelligence, and interaction. This calls forth the need for the development of additional trust measurement tools specific to the changing HRI environment.

This chapter summarizes research that was conducted to produce a reliable and validated subjective measure: the *Trust Perception Scale-HRI* (see also Schaefer 2013). The goal of this research was to design a subjective tool specific to the measurement of human-robot trust that could be expressed as an overall percentage of trust. In addition, this scale was designed to effectively measure trust perceptions over time, across robotic domains, by individuals in all the major roles of HRI (operator, supervisor, mechanic, peer, or bystander, as defined by Scholtz 2003), and across various levels of system autonomy and intelligence (see also Beer et al. 2014). To ensure that this new scale was valid, each part of scale development was constructed using the widely-accepted procedures discussed in DeVellis (2003) and Fink (2009). These procedures followed the protocol of large item pool creation, statistical item pool reduction, content validity assessment, and task-based validity testing.

10.2 Creation of an Item Pool

The first step in creating the *Trust Perception Scale-HRI* was to create an Item Pool. An Item Pool is a collection of relevant phrases or items that are associated with trust development. To meet this end, over 700 articles in the areas of human-robot trust, human-automation trust, and human-interpersonal trust were reviewed and analyzed. Theoretical, qualitative, and quantitative relationships were recorded. Potential items were then organized in relation to the *Three Factor Model of Human-Robot Trust* (Hancock et al. 2011). This model was then updated to incorporate potential antecedents of trust (see Fig. 10.1). Specific items to be included in the initial Item Pool were first drawn from these large scale literature reviews.

One major trust-specific finding from these reviews was the importance of design as it related to the robot’s physical form and functional capability. While some research had focused on the functional capabilities, limited experimental study had been conducted specifically related to the impact of robot form on trust development. Therefore, two initial experiments were conducted to assess this gap in the literature and further develop the initial Item Pool.

The purpose of the first study was to determine the relationship between physical form and trustworthiness, devoid of any direct information regarding the functional capabilities of the system. One hundred sixty-one participants rated 63 images of real-world industry, military, medical, service, social, entertainment, and therapy robots. These ratings included the degree to which the robot was perceived to be a machine, a robot, and an object, as well as its perceived intelligence (PI), level of

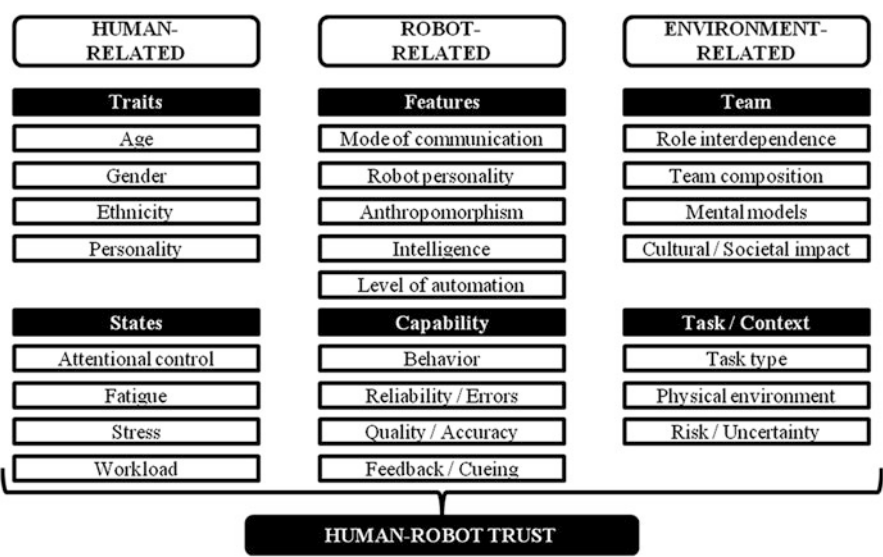


Fig. 10.1 Updated *Three Factor Model of Human-Robot Trust* following an extend literature review of trust in the interpersonal, automation, and robot domains

automation (LOA), trustworthiness, and the degree to which the participant would be likely to use or interact with the robot.

A multiple regression correlation analysis with stepwise entry of variables was conducted to determine the factors that predicted trustworthiness from perceived robot form alone. This was achieved by regressing trustworthiness onto human-related factors (gender, race, age, year in school), personality traits (agreeableness, extroversion, conscientiousness, intellect, neuroticism), negative attitudes toward robots (negative attitudes toward emotions in interactions, negative social influence, and negative situational influence), as well as self-report items of robot form (perceived intelligence, perceived level of automation (LOA), robot classification). The final model included *perceived intelligence (PI)*, *robot classification (RC)*, and *negative social influence (SI)* as predictors of *trustworthiness*, $\hat{Y} = 0.825 + 0.651(PI) + 0.256(RC) - 0.164(SI)$. It accounted for a significant R^2 of 45.1 % of the variance, $F(3|156) = 42.70$, $p < 0.001$.

These results suggested that preconceived ideas regarding the level of intelligence of a robot are form-dependent and assessed prior to interaction, in much the same way as one individual will assess another individual as a potential teammate. Further, *negative social influence* (e.g., capabilities, functions, etc.) plays a key role in expectation-setting similar to stereotypes of human teammates. Overall, the results of the above-mentioned study provided support that physical form is important to the trust that develops prior to HRI (for additional findings see also Schaefer et al. 2012).

The follow-up study was designed to identify which perceived robot attributes could impact the trustworthiness ratings. Robot attributes were assessed through a subset of the Godspeed questionnaire (Bartneck et al. 2009), a standardized measurement tool for HRI for interactive robots, specifically looking at items related to anthropomorphism (Powers et al. 2007), animacy (Lee et al. 2005), likeability (Monahan 1998), and perceived intelligence (Warner and Sugarman 1996). Over 200 participants rated a subset of the previous study's stimuli (two that were previously rated low on the robot classification scale, two that were rated high on the robot classification scale, and 14 that had diverse ratings on the robot classification scale). As anticipated, there was a significant relationship between how individuals rated the robot image on the robot classification scale and their perceived level of trustworthiness in the robot, $r(2910) = 0.307$, $p < 0.001$. The higher the rating of a robot to actually be classified as a robot, the more likely it was to be rated as trustworthy. The main purpose of this study was to determine if specific attributes could be identified to account for this relationship. Overall, results showed that each robot had different attributes that were important to classification. Therefore, it was decided to include all attribute items in the initial Item Pool.

Following the literature review and the two studies mentioned above, a full review of previously developed and referenced trust scales in the robot, automation, and interpersonal domains were reviewed to refine the items. This resulted in a review of 51 new scales (with a total of 487 trust items), 22 adapted versions of previously developed scales, and 13 previously developed scales (see also Table 10.1).

Table 10.1 Number of trust scales and trust items reviewed

Number of trust scales assessed	Robot	Automation	Interpersonal
Created new scales	9	30	12
Minimum number of items	1	1	1
Maximum number of items	45	31	29
Adapted previous scales	5	14	3
Previously developed scales	2	9	2
Scale were not discussed	2	11	4

	Strongly Disagree	Disagree	Slightly Disagree	Neutral	Slightly Agree	Agree	Strongly Agree
Most robots make poor teammates.	O	O	O	O	O	O	O
Most robots possess adequate decision-making capability.	O	O	O	O	O	O	O
Most robots are pleasant towards people.	O	O	O	O	O	O	O
Most robots are not precise in their actions.	O	O	O	O	O	O	O

Fig. 10.2 Example items included in the initial Item Pool

The final Item Pool resulted in the creation of 156 initial items. Between two and four items were created for each antecedent, representing equal number of positively and negatively worded (or opposite related) items. Initial scale items were written out as full sentences and referred to a general statement regarding “most robots” on a 7-point Likert-type scale (see Fig. 10.2).

10.3 Initial Item Pool Reduction

The second step in the scale development procedure was to reduce the size of the initial Item Pool using statistical procedures. These procedures began with a Principal Component Analysis (PCA) to identify potential groupings of items, as well as items that were not included in the groupings. Secondary analysis was conducted using paired samples *t*-tests to determine if the positively and negatively worded items were equal and thus could be reduced from the initial Item Pool.

10.3.1 Experimental Method

One hundred fifty-nine undergraduate students (65 males, 94 females) from the University of Central Florida took part in this study via online participation (SurveyMonkey.com). Following informed consent, participants completed the 156 randomized initial trust items. Participants then completed the demographics questionnaire that included gender, age, a mental model question, and prior experience questions. The study took approximately 30 min to complete. Participants’ prior experience with robots was assessed to understand previous exposure to robotic technologies. Prior experience has been shown to be related to how an individual forms a mental model of the robot and anticipates future HRI. As expected, the sample population had prior exposure to media representations (N = 156); some minor interaction with real-world robots (N = 36); and some opportunity to control (N = 34) or build (N = 11) a real-world robot during school or club related requirements. Table 10.2 presents results of these questions.

To assess the participants’ mental model of a robot, they were asked to describe what a robot looks like with an open-ended question. Mental models refer to structured, organized knowledge that humans possess which describe, explain, and predict a system’s purpose, form, function, or state (Rouse and Morris 1986). The responses were coded into categories (see Table 10.3). Seventeen participants directly referenced specific robots from movies or television (e.g., R2D2, C-3PO, iRobot, AI, and Terminator; N = 14); the video game Mass Effect 3 (N = 1); real-world military robots (e.g., Predator, N = 2); and a robotic arm (N = 2).

Table 10.2 Participants prior experience with robots

Prior experience questions	Yes (%)	No (%)
Have you ever watched a movie or television show that includes robots?	98	2
• 1–5 shows (N = 87)		
• 6–10 shows (N = 31)		
• Over 10 shows (N = 18)		
Have you ever interacted with a robot?	23	77
• Museum or theme park animatronics (N = 5)		
• Toys such as Furby (N = 8)		
• Robot vacuum (N = 2)		
• Classroom robots or Battlebots (N = 8)		
• Everyday items such as cell phone, computer, ATM, or Xbox (N = 12)		
• Unclassified (N = 1)		
Have you ever built a robot?	7	93
• Classroom or robotics club robots		
Have you ever controlled a robot?	21	79
• Teleoperation or remote control (N = 21)		
• Speech, Gesture, Commands (N = 3)		
• Computer programmed (N = 6)		

Table 10.3 Coding categories of mental model of a robot

Coding description	N	%
Machine-like (machine, metallic, silver)	121	76.1
Human-like (human-like or specific human features)	49	30.8
Varied (multiple descriptions or ranges of robots)	28	17.6
Tool	4	2.5
Task, Function, or Interaction	30	18.9
Internal Form: Computer, electronics, wires, buttons	25	15.7
External Form: shape, size, rigid, durable	34	21.4
Capabilities: movement	33	20.8
Communication: language	7	4.4
Other (helpful, intelligent, cameras, robot, alien)	7	4.4

10.3.2 Experimental Results

All data were analyzed using IBM SPSS v.19 (SPSS 2010), with an alpha level set to 0.05, unless otherwise indicated. These findings were important as they provided potential cause as to whether to retain or reject specific items from the initial Item Pool.

PCA was performed on the 156 initial trust items. Extraction was used to identify 43 components (using the Kaiser Criterion of Eigenvalue >1 for truncation), accounting for 79.63 % of the variance. Following review of the scree plot, four components were retained. The un-rotated solution was subject to orthogonal varimax rotation suppressed below |0.30|. In the rotated model, the four components accounted for 30.64 % of the variance. In looking at the loadings in the Rotated Component Matrix, 22 items with high loadings (>0.60) were located in Component 1. Based on the loadings of trust items on each of the four components, interpretations can be made about the factors themselves. Component 1 seemed to represent performance-based functional capabilities of the robot. Component 2 seemed to represent robot behaviors and communication. Component 3 may represent task or mission specific items. Finally, Component 4 seemed to represent feature-based descriptors of robots. These components supported the theory addressed by the descriptive *Three Factor Model of Human-Robot Trust* (first described by Hancock et al. 2011). Following PCA, 26 items were considered for immediate removal from the Item Pool.

Means, standard deviations, normality (skewness and kurtosis), correlations, z-scores, and paired samples *t*-tests were conducted to further assess items for retention or removal. To be retained in the scale, items should retain normality. Therefore, 62 items with significant skew and 20 items with significant kurtosis were considered for removal from the Item Pool. In addition, paired samples *t*-tests were conducted on all of the paired items (positive and negatively worded items) to determine if they were interchangeable, thus reducing the item pool. The results of

this assessment resulted in 39 paired items that were found to be not significantly different from each other. These results provided a rationale for reducing the scale by an additional 39 items.

Even though some elements might have been considered for removal, the following items were retained for subject matter expert (SME) review due to their importance to trust theory: move quickly, move slowly, require frequent maintenance, autonomous, led astray by unexpected changes in the environment, work in close proximity with people, possess adequate decision-making capability, make sensible decisions, openly communicate, and communicate only partial information. Following the various statistical assessments (PCA, normality assessment, and paired samples *t*-tests), the Item Pool was reduced from 156 items to 73 items.

10.3.3 Key Findings and Changes

Two major changes were made to the scale following this study. First, there were some potential issues that arose with the wording of the items. Two main types of item formation were included in the above version of the scale. Items either began with “Most robots” or “I.” This may have impacted the factor creation. Therefore, all items were reduced to a single word or short phrase prior to subject matter expert (SME) review. Secondly, the scale was modified from a 7-point Likert-type scale to a percentage scale with 10 % increments. The decision to make this change in the scale was related to larger purpose to develop a scale that provided a trust rating from no trust (0 %) to complete trust (100 %). This change was supported by research, especially in the interpersonal and e-commerce domains that suggest trust and distrust are viewed as related but separate constructs with differing effects on behavior, consequences, and outcomes (Lewicki et al. 1998; McKnight et al. 2004; Wildman 2011; Wildman et al. 2011).

10.4 Content Validation

The third step in the process to create a reliable and valid subjective scale was content validation. In this step, the goal was to survey SMEs in the area of trust and robotics in order to determine if each item should be retained or removed from the Item Pool. This two-phase semantic analysis included item relevance (content validation) using the protocols described by Lawshe (1975), and the identification of the hypothetical range of differences (e.g., no trust and complete trust differences) for each item.

Table 10.4 Years of experience for the subject matter experts

SME	Robot design	Robot operator	Robot research	HRI	Automation design	Automation research	Trust research
SME 1	0	0	8	8	0	8	0
SME 2	5	4	3	0	0	0	0
SME 3	4	0	4	0	2	2	0
SME 4	7	0	7	0	0	0	0
SME 5	4	8	8	8	0	0	0
SME 6	0	0	0	0	0	7	7
SME 7	11	0	11	11	0	0	3
SME 8	0	0	0	8	0	8	6
SME 9	0	0	4	0	0	4	0
SME 10	7	0	7	0	0	0	0
SME 11	0	0	10	10	20	30	15

All results are reported in years of experience

10.4.1 Experimental Method

Eleven SMEs were included from the United States Army Research Laboratory, United States Air Force Research Laboratory, and faculty members from university research laboratories. All SMEs were considered experts in the fields of trust research, robotics research, or HRI. Table 10.4 provides the SME’s years of experience across a variety of robot, automation, and research topics.

SMEs were contacted via email. Upon agreement to participate, they were provided a link to complete an online survey. All data for this experiment were collected through an online tool (SurveyMonkey.com). SMEs were provided background information, purpose, and a brief review of trust theory prior to beginning the multi-part study. In Part 1, SMEs were provided background information, purpose, and a brief review of trust theory prior to beginning the multi-part study. In Part 1, SMEs completed an expertise questionnaire. In Part 2, SMEs were given instructions to complete the 73 item Trust Scale with the instructions “Please rate the following items on how a person with little or no trust in a robot would rate them.” In Part 3, SMEs were given instructions to complete the 73 item Trust Scale with the instructions “Please rate the following items on how a person with complete trust in a robot would rate them.” All items in Part 2 and Part 3 were randomized. Part 4 was the Content Validation questionnaire based on Lawhe (1975) content analysis protocols. SMEs rated each item on a 3-point Likert-type scale as either “extremely important to include in scale,” “important to include in scale,” or “should not be included in scale.” SMEs could also mark if they felt an item was domain specific (e.g., military robotics, social robotics, etc.). A comment box was available to provide any clarification about why they rated the item a specific way, to provide additional recommendations to the scale design, or to suggest items that may be missing from the scale. The total survey took approximately 30 min to complete.

10.4.2 Experimental Results

Items were analyzed using the Content Validity Ratio developed by Lawshe (1975). The Content Validity Ratio (CVR), depicted in Eq. 10.1, is a commonly used method of analyzing scale items (see also Yagoda and Gillan 2012). The CVR equation was derived from a 3-point Likert scale (1 = Should not be included in scale, 2 = Might be important to include in scale, and 3 = Extremely important to include in scale).

CVR = (n_e - N/2) / (N/2) (10.1)

CVR = Content Validity Ratio
n_e = Number of SMEs indicating that an item is Extremely Important
N = Total number of SMEs

Lawshe’s protocol suggested that 11 SMEs with a criterion set to 0.59 are needed to ensure that the SME agreement is unlikely to be due to chance. The formula yielded values ranging from +1 to -1. Positive values indicated that at least half of the SMEs rated the item as Extremely Important. Table 10.5 reports the CVR values for the 14 items recommended by the SMEs.

CVR values were also calculated for the items that were rated as “important to include in the scale.” This resulted in 37 additional items to consider for inclusion in the finalized scale. The scores from the hypothetical range of differences were used to further evaluate these 37 items. The hypothetical range of differences was assessed from the SMEs completion of the Trust Scale in Part 2 and Part 3 of the study. Paired samples *t*-tests were conducted to identify the hypothetical range of differences (see Table 10.6).

Table 10.5 The 14 items recommended by SMEs as “Extremely Important”

Item	CVR values
1. Function successfully	1.00
2. Act consistently	1.00
3. Reliable	1.00
4. Predictable	1.00
5. Dependable	1.00
6. Follow directions	0.82
7. Meet the needs of the mission	0.82
8. Perform exactly as instructed	0.82
9. Have errors ^a	0.82
10. Provide appropriate information	0.82
11. Malfunction ^a	0.64
12. Communicate with people	0.64
13. Provide feedback	0.64
14. Unresponsive ^a	0.64

CVR ≥ 0.59
^aRepresents reverse coded items

Table 10.6 The 37 “Important Items” separated by retained and removed items

	CVR	Complete trust		No trust		Range	
		Mean	SD	Mean	SD	<i>t</i>	<i>p</i>
<i>Items retained</i>							
1. Operate in an integrated team environment	1.00	75.00	21.73	28.00	25.73	3.62	0.006
2. Autonomous	1.00	69.09	20.23	38.89	23.15	3.18	0.013
3. Good teammate	0.82	87.27	11.04	11.82	11.68	16.60	<0.001
4. Performs a task better than a novice human user	0.82	69.09	20.71	24.55	24.23	6.69	<0.001
5. Led astray by unexpected changes in the environment	0.82	71.00	19.12	26.00	15.78	5.78	<0.001
6. Know the difference between friend and foe	0.82	71.00	27.67	14.55	16.95	5.89	<0.001
7. Make sensible decisions	0.82	84.00	11.74	21.00	20.79	8.62	<0.001
8. Clearly communicate	0.82	83.00	11.60	19.00	15.24	12.30	<0.001
9. Warn people of potential risks in the environment	0.82	83.00	11.60	23.00	18.29	7.75	<0.001
10. Incompetent	0.82	85.45	14.40	39.00	33.15	5.24	0.001
11. Possess adequate decision-making capability	0.82	71.11	16.91	20.00	21.60	5.57	0.001
12. Are considered part of the team	0.82	79.00	12.87	35.00	31.36	3.36	0.010
13. Will act as part of the team	0.82	74.00	19.55	30.00	29.06	3.28	0.010
14. Perform many functions at one time	0.82	68.00	23.48	36.00	25.47	4.40	0.002
15. Protect people	0.82	77.00	24.52	23.00	22.14	3.92	0.003
16. Openly communicate	0.82	80.00	15.81	34.00	28.36	3.79	0.005
17. Responsible	0.82	66.36	30.42	27.27	34.67	2.76	0.020
18. Built to last	0.82						
19. Work in close proximity with people	0.82	65.00	19.58	35.56	26.03	2.34	0.047
20. Supportive	0.64	66.00	16.47	18.00	11.35	8.67	<0.001
21. Work best with a team	0.64	71.00	18.53	34.00	28.36	3.41	0.008
22. Tell the truth	0.64	86.36	21.57	46.00	31.69	2.90	0.018
23. Keep classified information secure	0.64	84.00	15.06	55.45	36.43	2.69	0.025
24. Require frequent maintenance	0.64	74.00	20.66	44.00	28.75	2.37	0.042
<i>Items removed from scale</i>							
25. Responsive	1.00	84.00	9.66	32.22	21.67	7.50	<0.001
26. Poor teammate	0.82	84.55	9.66	45.45	36.43	3.64	0.005
27. Are assigned tasks that are critical to mission success	0.82	58.00	32.25	25.00	35.67	1.84	0.098

(continued)

Table 10.6 (continued)

	CVR	Complete trust		No trust		Range	
		Mean	SD	Mean	SD	<i>t</i>	<i>p</i>
28. Communicated only partial information	0.82	53.00	28.30	35.00	28.77	1.03	0.331
29. Instill fear in people	0.73	83.64	18.59	50.91	34.48	3.13	0.011
30. Likeable	0.64	60.00	33.54	14.00	10.75	4.61	0.001
31. Easy to maintain	0.64	60.00	33.54	35.00	31.71	2.89	0.018
32. Responsible for its own actions	0.64	64.00	32.04	32.73	36.63	2.22	0.054
33. Given complete responsibility for the completion of a mission	0.64	59.00	37.55	19.00	37.84	2.16	0.059
34. Monitored during a mission	0.64	49.00	29.61	20.00	28.67	2.08	0.067
35. Are considered separate from the team	0.64	68.00	24.40	36.00	32.39	1.92	0.091
36. Difficult to maintain	0.64	70.00	15.63	50.00	34.64	1.60	0.143
37. Work best alone	0.64	48.00	23.94	41.00	33.48	0.69	0.506

Twenty-four items were retained for further scale validation. The remaining 13 items were removed from the scale for the following reasons: non-significant findings on the paired samples *t*-tests for eight items, and SMEs comments assisted in the removal of the remaining five items. The first comment was a general comment stating that some items (e.g., easy/difficult to maintain) were repetitive in nature. To address this comment, only one of the repetitive items was included in the revised scale. *Responsive*, *easy to maintain*, and *poor teammate* were removed. In addition *instill fear in people* was removed due to its nature of distrust more than trust. Finally, the item *likeable* was considered to be too general an item for the scale. An additional comment suggested that two items *are given complete responsibility for the completion of the mission*, and *are assigned tasks that are critical to mission success* represented situational factors that may be a separate issue from trust. Even though CVR analysis revealed that SMEs felt that they might be important items to include in the scale, their responses to the theoretical range of scores did not show a significant change in trust. This added support for the SMEs' comments recommending removal of these two items.

Twenty-two items did not meet the CVR criterion and were reviewed for removal from the scale. Of these 22 items, the four items regarding movement (move quickly, move slowly, mobile, move rigidly) were removed. To further support this decision, there was a comment from a SME suggesting that trust and speed were orthogonal. SMEs further recommended that three items representing robot personality (kind, caring, have a relationship with their human users or operators) were recommended for removal. An additional nine of these items (human-like, fake, alive, dead, offensive, organic, ignorant, apathetic, and make decisions that affect me personally) were removed from the scale due to the lack of change in scores on the theoretical range of responses. However, out of those 22 items that did not meet the CVR criterion, a number of SMEs felt that some items may have particular importance

to trust development as robots advance further into socially relevant relationships. Therefore, the following four items were retained in the scale: friendly, pleasant, conscious, and lifelike.

Semantic analysis of the trust scale items reduced the scale from 73 items to 42 items. It further identified 14 items that were extremely important to trust measurement, with an additional 24 items that could be important to trust measurement. Four domain specific items (friendly, pleasant, conscious, and lifelike) were also retained on the scale based on SME recommendations. No additional items or item revisions were recommended by the SMEs. In addition, no changes to the scale design were recommended.

10.5 Task-Based Validity Testing: Does the Score Change Over Time with an Intervention?

Trust has typically been measured following an interaction. However, trust is dynamic in nature, as ongoing interactions and relational history continuously influence trust levels at any given point in time. Consequently, trust before, during, and after an interaction may not be identical. Further, trust in the same partner will likely change over time as the relationship progresses (Bloomqvist 1997). Research in the areas of automation (see Merritt and Ilgen 2008), as well as interpersonal trust (see McAllister 1995), recommended that trust should be measured at multiple times, specifically before and after the task or interaction. Within the *Three Factor Model of Human-Robot Trust*, pre-interaction measurement has been used to identify initial trust perceptions that are influenced by human traits, robot features, and the individual's perception of the environment and perceived robot capabilities. Post-interaction measurement was used to identify changes in trust related to human states and trust perceptions following interactions. Obtaining the most reliable and accurate reflection of the dynamic nature of trust in an interaction may necessitate measuring trust multiple times.

In order to examine this concept of dynamic trust, an experiment was designed to assess the 42 Item Trust Scale's capability to measure changes in perceived trust. Computer-based simulation was used to develop a monitoring task specific to a "screen the back" scenario for a Soldier-robot team (Army Research Laboratory 2012). Trust, as measured using the 42 Item Trust Scale, was assessed both pre-interaction, and post-interaction in the high trust condition (the robot provided 100 % reliable feedback on target detection), as well as post-interaction in the low trust condition (the robot provided 25 % reliable feedback). It was hypothesized that there are mean differences in trust that occur over time, with respect to changes in robot reliability. More specifically, trust will increase from pre-interaction to after experiencing a 100 % reliable interaction; trust will decrease after first experiencing a 100 % reliable interaction and then experiencing a 25 % reliable interaction.

10.5.1 *Experimental Method*

Participants included 81 undergraduate students (25 males, 56 females; $M = 22.57$ years, $SD = 3.95$) from an undergraduate psychology course in Science and Pseudoscience at the University of Central Florida. Participants had varied backgrounds with respect to robot familiarity. All participants previously watched movies or television shows that incorporated robots: 1–5 shows ($N = 30$), 6–10 shows ($N = 9$), and over 10 shows ($N = 42$). Forty-four participants previously interacted with robots, ranging from the Roomba™ vacuum cleaner to a bomb disposal robot. Participants also reported previously controlling robots through a variety of modalities: voice ($N = 11$), game controller ($N = 24$), gestures or pictures ($N = 3$), or a radio control (RC) system ($N = 35$). In addition, five participants previously built robots for class-based projects.

All materials were administered through paper and pencil versions. The Negative Attitudes toward Robots Scale (NARS; Nomura et al. 2004), and demographics questionnaire were included to identify potential individual difference ratings. The NARS has three subscales: NARS_S1 represents negative attitudes toward situations of interaction with robots; NARS_S2 represents negative attitudes toward social influence of robots; and NARS_S3 represents negative emotions in interactions with robots. Trait-based trust was measured through the Interpersonal Trust Scale (ITS; Rotter 1967). State-based trust was measured through the 42 item Trust Scale, administered pre- and post-interaction. Participants' states were not assessed in this experiment due to the nature of the task (monitoring only).

All HRI scenarios were developed using the Robotic Interactive Visualization & Experimental Technology (RIVET) computer-based simulation system developed by General Dynamics Robotic Systems (GDRS; Gonzalez et al. 2009), in collaboration with the US Army Research Laboratory. RIVET uses an adapted Torque Software Development Kit (SDK) development and runtime environment through a Client/Server networking model. Development of the virtual environment (VE) was accomplished through a TorqueScript language, similar to C++. The VEs used a base environment developed previously for Army research activities. This included the layout of the physical environment (e.g., ground, roadways, buildings, and lighting), as well as inclusion of the Soldier and terrorist non-player characters. Task-specific customization of the environment was accomplished through scripting syntax. Examples of specific customization included entering objects, obstacles, and creation of paths.

An experimental scenario was created using the RIVET computer-based simulation system of a Soldier-Talon™ robot team completing a “screen the back” mission (Army Research Laboratory 2012). Participants monitored the Talon robot as it navigated to the back of the building, repositioned behind a barrel, monitored the back of the building for human targets, and provided a speech-based response when a target was detected. A single video was created from the camera-view on the Talon™ robot. The video of the simulation was created using FRAPS® real-time video capture and benchmarking program with a 30 frame rate/s .avi file.

The .avi file was converted into the .mp4 file format to add auditory feedback. The robot stated “target detected” in a male computer synthetic voice. There were eight possible targets included in each scenario. No false alarms were included.

The study was conducted in a single session. Participants were provided a copy of the informed consent while the experimenter read it aloud. Participants then viewed an image of the Talon robot and completed the 42 Item Trust Scale (Time 1; pre-interaction). Next, participants were instructed about the human-robot task they were about to monitor. Following Video 1, participants completed the 42 Item Trust Scale (Time 2; post-interaction, 100 % reliable feedback). Participants then received the second task instructions, monitored Video 2, and completed the 42 Item Trust Scale (Time 3; post-interaction, 25 % reliable feedback). Following these monitoring tasks, participants completed the ITS, NARS, and demographics questionnaire. The study took approximately 90 min in its entirety.

10.5.2 Experimental Results

All data were analyzed using IBM SPSS Statistics v.19 (SPSS 2010), with an alpha level set to 0.05, unless otherwise indicated.

10.5.2.1 Individual Item Analysis

The first step of analysis was to determine if each item changed over time. A one-way within-subjects repeated measures analysis of variance was conducted for each of the 42 items of the Trust Scale. The factor was “time” and the dependent variable was the individual score. Thirty-four items showed a significant mean difference for the condition of “time.” Using Post-hoc analysis, six items were found to only have significant differences between Time 1 (Pre-Interaction) and Time 2 (post-interaction). These items included: *lifelike*, *perform many functions at one time*, *friendly*, *know the difference between friend and foe*, *keep classified information secure*, and *work in close proximity to people*. These results may have occurred due to a significant change in the mental model from pre- to post-interaction; therefore, the items were retained in the scale. Confidence interval analysis was conducted on the remaining two items (*operate in an integrated team environment*, *built to last*) and showed no significant change between Time 1, Time 2, or Time 3. These items were removed from the scale. Additional analyses were conducted using the 40 retained items.

10.5.2.2 Trust Score Validation

First, a general score of trust was created for each of the three time periods. Following reverse coding of specific items, the 40 items were summed and divided

by 40 to formulate a score between 0 and 100. To assess the impact of time on trust development, a one-way within-subject repeated measures analysis of variance was conducted. The results indicated a significant effect of time, $F(2, 79) = 119.10$, $p < 0.001$, $\eta p^2 = 0.75$. Post-hoc analysis using the Fisher LSD revealed significant mean difference ratings for all three times the trust scale was administered. It revealed that trust was significantly greater in Time 2 (post-interaction, 100 % reliable condition) than Time 1 (pre-interaction) and Time 3 (post-interaction, 25 % reliable condition), thus supporting the hypothesis. In addition, mean trust scores at Time 1 were significantly greater than at Time 3 (see Fig. 10.3).

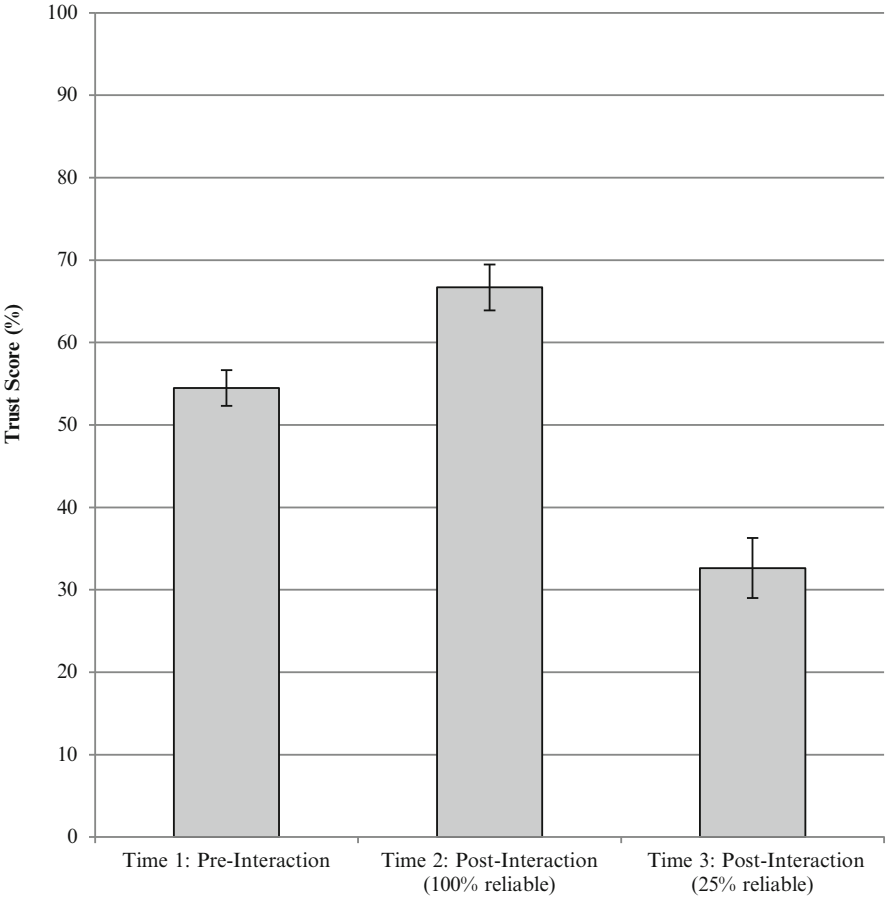


Fig. 10.3 Bar graph representing significant mean differences of trust over time, with 95 % confidence interval error bars

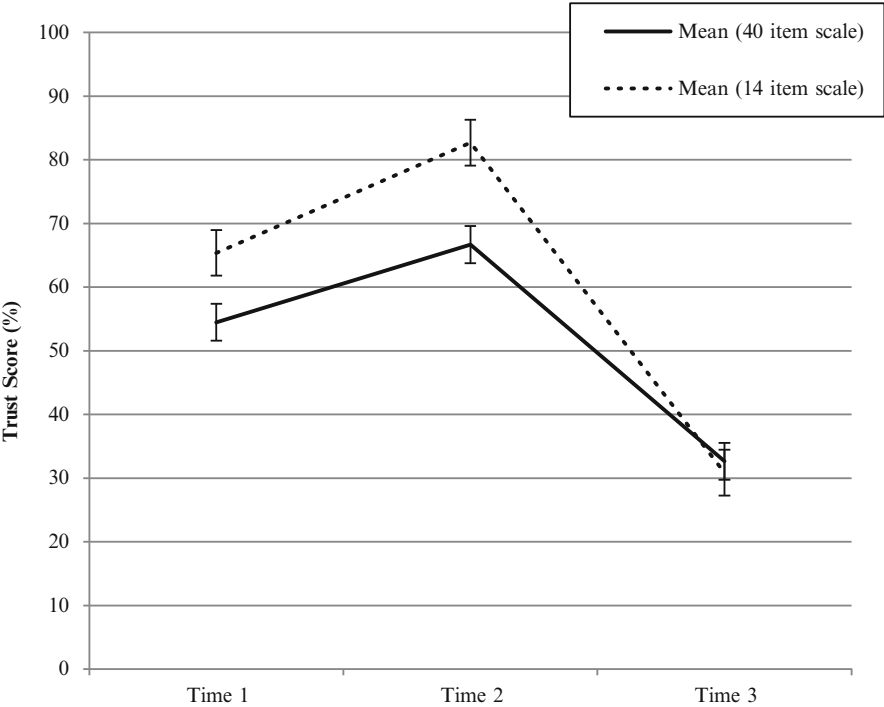


Fig. 10.4 Trust scores for the 40 item and the 14 item scale across time

10.5.2.3 40 Items Versus 14 Items

Additional analyses were also conducted to identify the differences between the 40 Item Trust Scale and the 14 Item SME recommended scale. A 2 Scale (40 items, 14 items) × 3 Time (Time 1, Time 2, Time 3) repeated measures analysis of variance was conducted (see Fig. 10.4).

Results showed a significant effect of Time, $F(2240) = 186.59$, $p < 0.001$, $\eta^2 = 0.609$; Scale $F(1240) = 273.61$, $p < 0.001$, $\eta^2 = 0.533$; and an interaction between Time and Scale, $F(2240) = 108.84$, $p < 0.001$, $\eta^2 = 0.476$. Review of the confidence intervals showed a significant difference between the scales at Time 1 and Time 2, but not Time 3. While findings revealed significant differences between the two scales, graphical representations showed similar patterns in the results. Taking into account both the individual analyses of each item measured over Time, as well as the comparative results of the two scales, it appeared that the total trust score of the 40 Item Trust Scale provided a finer level of granularity and thus a more accurate trust rating.

10.6 Task-Based Validity Testing: Does the Scale Measure Trust?

This study marked the final validation experiment for the 40 Item Trust Scale. It used a Same-Trait approach (Campbell and Fiske 1959) to validate that this scale measured trust and not an alternative construct. The same-trait was evaluated through a comparison of the developed 40 Item Trust Scale, and the well-established Checklist for Trust between People and Automation (Jian et al. 1998) trust in automation scale. Human-robot interaction was accomplished through computer-based simulation of a joint navigation task.

It was first hypothesized that there would be a strong positive correlation between the 40 Item Scale, the 14 Item SME selected subscale, and Checklist for Trust between People and Automation (Jian et al. 1998) trust scales. The second hypothesis was that the three change scores in the post-interaction conditions (20 % robot navigation errors—80 % robot navigation errors) for the types of trust scales (i.e., 40 item scale, 14 item scale, and Checklist for Trust) would not show significant mean differences. However, it was anticipated that the 40 Item and 14 Item scales would change from pre-interaction measurement to post-interaction measurement, as shown in the prior experiment.

10.6.1 *Experimental Method*

Twenty-one undergraduate students from the University of Central Florida (12 males, 9 females) participated in two Soldier-robot team-based computer simulations to provide the next level of task-based validity testing. Multiple scales were included to measure subjective trust, personality traits, demographics, and human states. Subjective trust was measured through the developed 40 Item Trust Scale. A partial measure represented by the 14 Items recommended by SME's was also assessed. The well-established Checklist for Trust between People and Automation (Jian et al. 1998) was included for Same-Trait analysis. Items including the word 'automation' were adapted to 'robot.' The Interpersonal Trust Scale (Rotter 1967) was also included. The 7-point Mini-IPIP personality assessment (Donnellan et al. 2006) was used to measure the Big 5 personality traits: agreeableness, extraversion, intellect, conscientiousness, and neuroticism. The Dundee Stress State Questionnaire (DSSQ; Matthews et al. 1999) was included to measure human states (i.e., mood state, motivation, workload, and thinking style) before and after a task.

The virtual environment (VE) for the present procedure was developed in RIVET and used a base environment of a Middle Eastern town developed by GDRS in collaboration with the US Army Research Laboratory. Independent task-specific customization of the physical environment was accomplished through scripting syntax. Specific customization included entering objects, obstacles, and creation of paths, etc. Scripting files were created for both the training session

and the task conditions. The task was to assist an autonomous robot from a set location to a rendezvous point. Participants controlled a Soldier avatar throughout the Middle Eastern town using a keyboard and mouse interface. It was possible that the robot could become stuck on an obstacle and required the participant's assistance. Participants could move certain obstacles out of the way of the robot by simply walking into the obstacle. The mission ended when both the Soldier and robot reached the rendezvous location. In Simulation A, the robot autonomously navigated around four out of the five obstacles. In Simulation B, the robot only navigated around one of the obstacles. Each simulation was approximately 1 min in length. The order of simulation presentation was counterbalanced and determined prior to participation.

Following completion of informed consent, participants completed three questionnaires: the demographics questionnaire, the Mini-IPIP personality inventory, and the ITS. Participants were then shown a picture of the Talon robot and completed the 40 Item Trust Scale, the Checklist for Trust between People and Automation, and the DSSQ to acquire baseline information. Participants then completed the two simulated tasks, followed by completion of the two trust scales and the post-task DSSQ following each task. The simulated tasks were recorded using FRAPS® real-time video capture and benchmarking program with a 30 frame rate/s .avi file. Video was recorded from the Soldier character's perspective. It was saved with a unique identifier to maintain participant confidentiality. The entire study took approximately 1 h to complete.

10.6.2 Experimental Results

All data were analyzed using IBM SPSS Statistics v.19 (SPSS 2010), with an alpha level set to 0.05, unless otherwise indicated. Initial analyses were conducted to assess changes in human states over time. Results demonstrated no significant difference in mood state or motivation subscales. The thinking style subscales of self-focused attention and concentration showed a significant difference between pre-interaction and post-interaction, but no difference between the two post-interaction conditions. A similar result was found for the thinking content subscale, task interference. Due to these findings, no additional analyses were conducted assessing human states. The Same-Trait methodology compared the developed 40 Item Trust Scale, the SME's recommended 14 Item Trust Scale, and the Checklist for Trust between People and Automation.

10.6.2.1 Correlation Analysis of the Three Scales

The first step in this validation was to identify the relationships between the three trust scales. In support of Hypothesis 1, significant positive Pearson correlations were found between all three scales (see Table 10.7).

Table 10.7 Same-trait trust scale correlations over time

	M	SD	1	2	3
<i>Pre-interaction trust</i>					
1. 40 Item Scale	60.60	13.58	1		
2. 14 Item Scale	72.45	12.70	0.829**	1	
3. Checklist for Trust between People and Automation	71.71	16.27	0.620**	0.745**	1
<i>Post-interaction (20 % errors)</i>					
1. 40 Item Scale	49.92	19.59	1		
2. 14 Item Scale	60.61	20.19	0.918**	1	
3. Checklist for Trust between People and Automation	71.54	16.22	0.857**	0.854**	1
<i>Post-interaction (80 % errors)</i>					
1. 40 Item Scale	46.70	22.43	1		
2. 14 Item Scale	57.42	24.93	0.934**	1	
3. Checklist for Trust between People and Automation	72.27	17.16	0.855**	0.852**	1

** represent significance at the .01 level

The second step in this validation process was to determine if there was a significant mean difference between the post-interaction change scores (20 % robot navigation errors—80 % robot navigation errors) for the three scales. A within-subjects repeated measures analysis of variance was conducted. In support of Hypothesis 2, there was not a significant mean difference between the post-interaction change scores for the three trust scales, $F(2,19) = 2.64$, $p = 0.097$. This result provided additional support that the developed scale measures the construct of trust.

10.6.2.2 Pre-post Interaction Analysis

Additional analyses were conducted to determine the differences between the pre-post interaction scores between the three scales. Paired samples t -tests were conducted to assess the change in trust pre-post interaction. A significant change in trust was found between the pre-post interaction trust measurement for the 40 Item Trust Scale, $t(40) = 3.87$, $p < 0.001$, and the 14 Item Trust Scale, $t(40) = 3.86$, $p < 0.001$. However, the Checklist for Trust between People and Automation trust scale did not change ($p = 0.932$). This finding suggested that the developed trust scale does indeed measure something additional to the previously developed trust scale.

10.6.2.3 Differences Across Scales and Conditions

To further explore these scale differences, a 3 Trust Scales (40 item, 14 item, and Checklist for Trust) × 3 Conditions (pre-interaction, 20 % robot error, and 80 % robot error) repeated measures analysis of variance was conducted. There was a main effect of scale, $F(2,59) = 105.16$, $p < 0.001$ $\eta^2 = 0.781$, but not condition ($p = 0.191$). Results are depicted in Fig. 10.5. Confidence interval analysis of the mean trust scores demonstrated that there was no significant difference between the three scales that were recorded pre-interaction. This finding suggested that all three scales provide similar trust scores prior to HRI. Further there were no significant differences between the 14 Item Trust Scale and the well-established Checklist for Trust between People and Automation (Jian et al. 1998). This finding is not surprising as the items from both the 14 Item Trust Scale and the Jian et al. scale referenced the capability of the system. The important finding was the significant differences between the 40 item Trust Scale and the Jian et al. scale during the two post-interaction conditions.

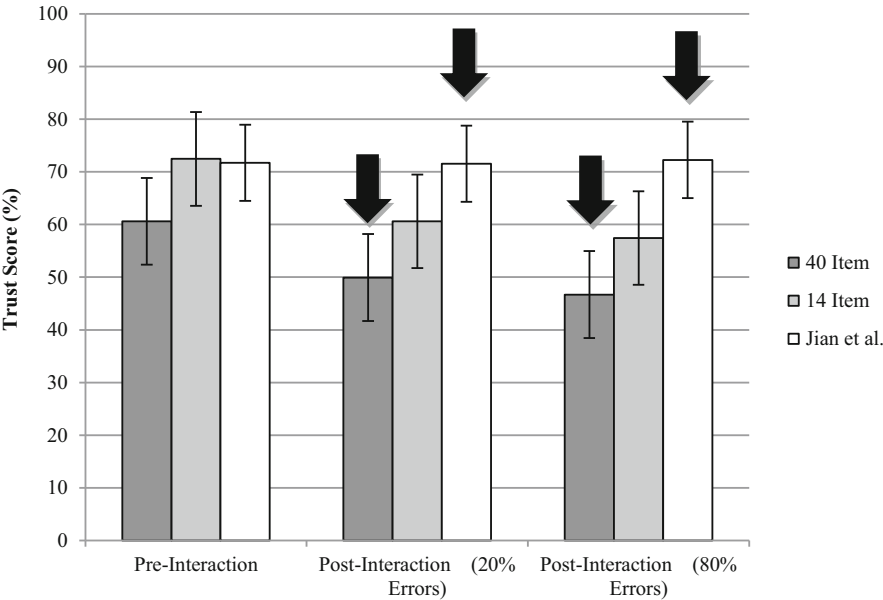


Fig. 10.5 Differences between the 40 Item, 14 Item, and Checklist for Trust between People and Automation (Jian et al. 1998) trust scales

10.6.3 Experimental Discussion

This study demonstrated that the developed trust scale assessed the construct of trust. In addition, it provided support for additional benefits of the developed 40 Item Trust Scale above and beyond previously used scales (i.e., Checklist for Trust between People and Automation; Jian et al. 1998). First, there were strong positive correlations between the three scales. Second, mean analysis showed significant differences between the 40 item and Jian et al. scale in the post-interaction conditions. Both the 40 Item and the 14 Item scales showed a significant change in trust from pre-interaction to post-interaction; however the Checklist for Trust did not change. This change in trust was mirrored in the previous study, and is supported by the trust theory. Therefore, it can be postulated that the developed Trust Scale accounted for the relationship between the change in mental models and trust development that occurs after HRI. In addition, findings from this study, together with the findings from the previous study provide support that the 40 Item Trust Scale had more accurate trust scores than both the 14 Item SME recommended scale and the Checklist for Trust scale.

10.7 Conclusion

The goal was to develop a trust perception scale that focused on the antecedents and measurable factors of trust specific to the human, robot, and environmental elements. This resulted in the creation of the 40 item *Trust Perception Scale-HRI* and the 14 item sub-scale. The finalized scale was designed as a pre-post interaction measure used to assess changes in trust perception specific to HRI. The scale was also designed to be used as post-interaction measure to compare changes in trust across multiple conditions. It was further designed to be applicable across all robot domains. Therefore, this scale can benefit future robotic development specific to the interaction between humans and robots.

10.7.1 The Trust Perception Scale-HRI

The scale provided an overall percentage score across all items. Items were preceded by the question “What percentage of the time will this robot . . .” followed by a list of the items. The finalized 40 item scale is provided in Table 10.8, and took between 5 and 10 min to complete.

Table 10.8 Finalized Trust Perception Scale-HRI

	0 %	10 %	20 %	30 %	40 %	50 %	60 %	70 %	80 %	90 %	100 %
<i>What % of the time will this robot be . . .</i>											
1. Considered part of the team	O	O	O	O	O	O	O	O	O	O	O
2. Responsible	O	O	O	O	O	O	O	O	O	O	O
3. Supportive	O	O	O	O	O	O	O	O	O	O	O
4. Incompetent ^a	O	O	O	O	O	O	O	O	O	O	O
5. Dependable ^b	O	O	O	O	O	O	O	O	O	O	O
6. Friendly	O	O	O	O	O	O	O	O	O	O	O
7. Reliable ^b	O	O	O	O	O	O	O	O	O	O	O
8. Pleasant	O	O	O	O	O	O	O	O	O	O	O
9. Unresponsive ^{a,b}	O	O	O	O	O	O	O	O	O	O	O
10. Autonomous	O	O	O	O	O	O	O	O	O	O	O
11. Predictable ^b	O	O	O	O	O	O	O	O	O	O	O
12. Conscious	O	O	O	O	O	O	O	O	O	O	O
13. Lifelike	O	O	O	O	O	O	O	O	O	O	O
14. A good teammate	O	O	O	O	O	O	O	O	O	O	O
15. Led astray by unexpected changes in the environment	O	O	O	O	O	O	O	O	O	O	O
<i>What % of the time will this robot . . .</i>											
16. Act consistently ^b	O	O	O	O	O	O	O	O	O	O	O
17. Protect people	O	O	O	O	O	O	O	O	O	O	O
18. Act as part of the team	O	O	O	O	O	O	O	O	O	O	O
19. Function successfully	O	O	O	O	O	O	O	O	O	O	O
20. Malfunction ^a	O	O	O	O	O	O	O	O	O	O	O
21. Clearly communicate	O	O	O	O	O	O	O	O	O	O	O
22. Require frequent maintenance ^a	O	O	O	O	O	O	O	O	O	O	O
23. Openly communicate	O	O	O	O	O	O	O	O	O	O	O
24. Have errors ^a	O	O	O	O	O	O	O	O	O	O	O
25. Perform a task better than a novice human user	O	O	O	O	O	O	O	O	O	O	O
26. Know the difference between friend and foe	O	O	O	O	O	O	O	O	O	O	O
27. Provide feedback ^b	O	O	O	O	O	O	O	O	O	O	O
28. Possess adequate decision-making capability	O	O	O	O	O	O	O	O	O	O	O
29. Warn people of potential risks in the environment	O	O	O	O	O	O	O	O	O	O	O
30. Meet the needs of the mission/task ^b	O	O	O	O	O	O	O	O	O	O	O

(continued)

Table 10.8 (continued)

	0 %	10 %	20 %	30 %	40 %	50 %	60 %	70 %	80 %	90 %	100 %
31. Provide appropriate information ^b	O	O	O	O	O	O	O	O	O	O	O
32. Communicate with people ^b	O	O	O	O	O	O	O	O	O	O	O
33. Work best with a team	O	O	O	O	O	O	O	O	O	O	O
34. Keep classified information secure	O	O	O	O	O	O	O	O	O	O	O
35. Perform exactly as instructed ^b	O	O	O	O	O	O	O	O	O	O	O
36. Make sensible decisions	O	O	O	O	O	O	O	O	O	O	O
37. Work in close proximity with people	O	O	O	O	O	O	O	O	O	O	O
38. Tell the truth	O	O	O	O	O	O	O	O	O	O	O
39. Perform many functions at one time	O	O	O	O	O	O	O	O	O	O	O
40. Follow directions ^b	O	O	O	O	O	O	O	O	O	O	O

^aRepresents the reverse coded items for scoring

^bRepresents the 14 item sub-scale items

10.7.2 *Instruction for Use*

When the scale is used as a pre-post interaction measure, the participants should first be shown a picture of the robot they will be interacting with or provided a description of the task prior to completing the pre-interaction scale. This accounts for any mental model effects of robots and allows for comparison specific to the robot at hand. For post-interaction measurement, the scale should be administered directly following the interaction. To create the overall trust score, 5 items must first be reverse coded. The reverse coded items are denoted in the above table. All items are then summed and divided by the total number of items (40). This provides an overall percentage of trust score.

While use of the 40 Item scale is recommended, a 14 Item subscale can be used to provide rapid trust measurement specific to measuring changes in trust over time, or during assessment with multiple trials or time restrictions. This subscale is specific to functional capabilities of the robot, and therefore may not account for changes in trust due to the feature-based antecedents of the robot. Trust score is calculated by first reverse coding the ‘have errors,’ ‘unresponsive,’ and ‘malfunction’ items, and then summing the 14 item scores and dividing by 14. The 14 items are marked in Table 10.8.

10.7.3 *Current and Future Applications*

This scale was developed to provide a means to subjectively measure trust perceptions over time and across robotic domains. In addition, it can be used by individuals in all the major roles of HRI: operator, supervisor, mechanic, peer, or bystander. Therefore, there are many potential avenues for future research using *Trust Perception Scale-HRI*. Current and near-term research studies highlight the expansion of human-robot trust antecedents, which were first described in the *Three Factor Model of Human-Robot Trust* (Hancock et al. 2011; Schaefer et al. 2014). These include the exploration of human-robot trust as it relates to: Soldier, bystander, and robot proximity in high-risk military tasking; transparent system communication and feedback for Soldier-robot teaming in high-risk environments; the development of natural language processing in human-robot teams; and dual-task engagement with an autonomous vehicle designed for on-base personnel transport. While a number of these studies are currently under review or in press, a few preliminary results are discussed below.

First, Sanders et al. (2014) used the *Trust Perception Scale-HRI* to assess the impact of the amount of communication feedback from the robot (constant, contextual only, minimal) and modality of information (visual, text, audio) on trust development. Their initial results found a greater increase in the change in trust (post-interaction minus pre-interaction) for a constant stream of information compared to contextual information only and minimal information across three different communication modalities (text, auditory, and visual) during a Soldier-robot team surveillance task clearing an area of weapons and locating civilians to be safely evacuated from a hostile zone. These researchers are continuing to use the *Trust Perception Scale-HRI* for future studies in transparent communication, human roles (team member, bystanders), and social dynamics including proxemics, as part of the US Army Research Laboratory's Robotics Collaborative Technology Alliance tasking related to determinants of shared cognition and social dynamics in future Soldier-robot teams.

Second, a recent study using the *Trust Perception Scale-HRI* was conducted where participants monitored a simulated and a live robot surveying an environment, locating an object, and touching the object (Schaefer et al. 2015). Results supported previous findings that individuals trust a reliable robot significantly more than the unreliable robot. Also, the results showed that the scale is effective for measuring trust in both simulated and live HRI experimentation.

A third area of on-going and near-term work using the *Trust Perception Scale-HRI* is exploring trust development with respect to the development of robotic passenger vehicles and transparent passenger user interfaces (Schaefer 2015). The design of this set of computer-based simulation studies is in line with the goals of the US Army Tank Automotive Research, Development and Engineering Center's ARIBO (Applied Robotics for Installation and Base Operations) project for alternative transportation options for on-base wounded Soldier transit (Marshall 2014). The ultimate goal is to provide a means for Soldiers to schedule an on-demand

autonomous robotic passenger vehicle to arrive and drive door-to-door to and from the Medical Barracks to the on-base medical facilities. The benefit of this work is to understand the levels of trust that will enhance usage and effective human-robot interaction. Results of this set of studies will provide additional insight into the impact of trust antecedents (e.g., transparency, cueing, human characteristics) on trust development, as well as explore how the trust relationship changes as the human role transitions from a driver, to a safety rider, to a supervisor external to the vehicle, as well as to a passenger. Initial findings advance current trust theory by demonstrating significant relationships with working memory capacity and coping style related to driving, distress, workload, and task performance (Schaefer & Scribner 2015). Future work on this project is two part: (1) understanding the effects of the availability of driver control interfaces (i.e., steering and speed control versus automation engage/disengage buttons) on usability, performance, and trust; and (2) exploring the effects of transparent user interfaces design on trust development and calibration for passengers.

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